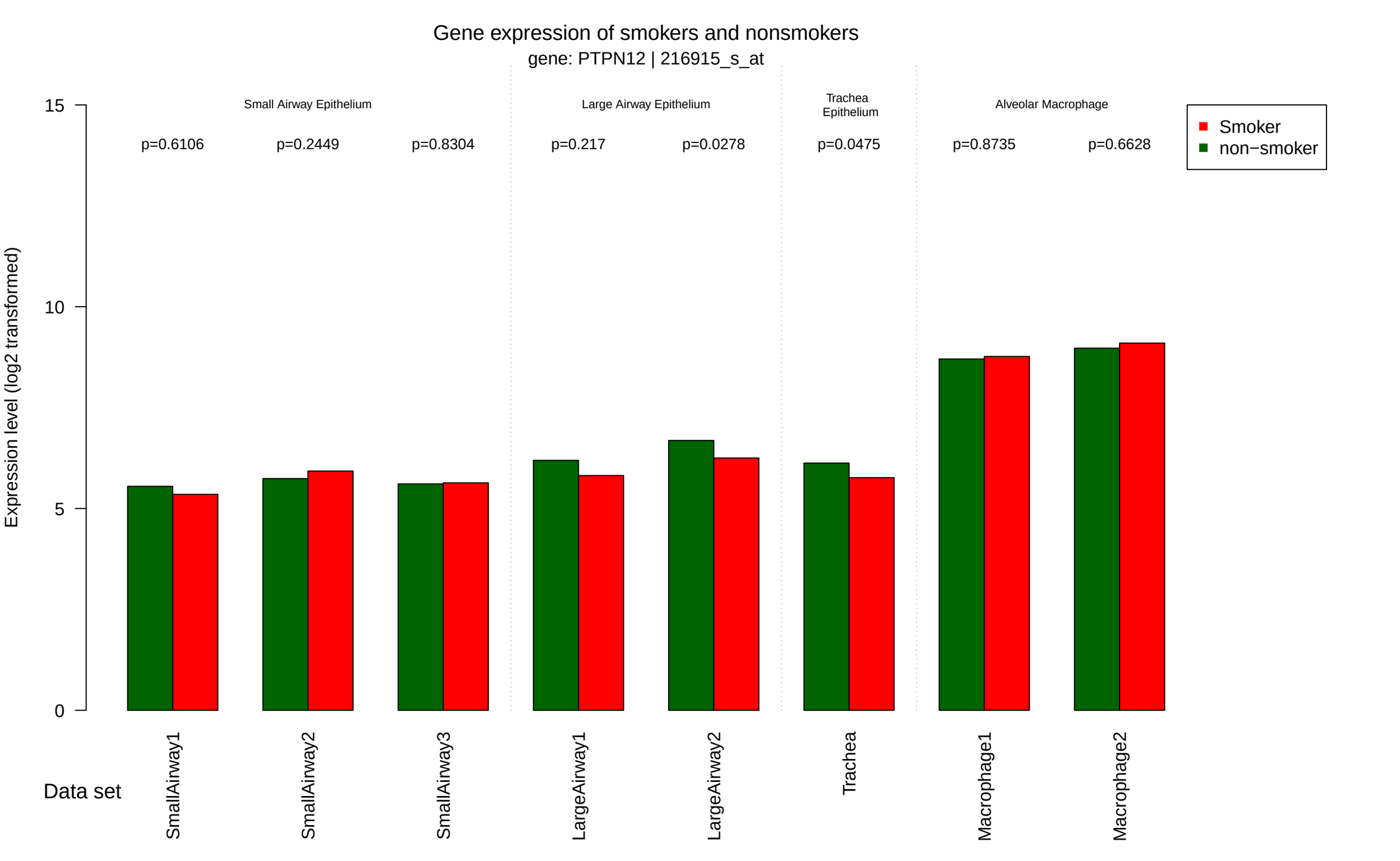
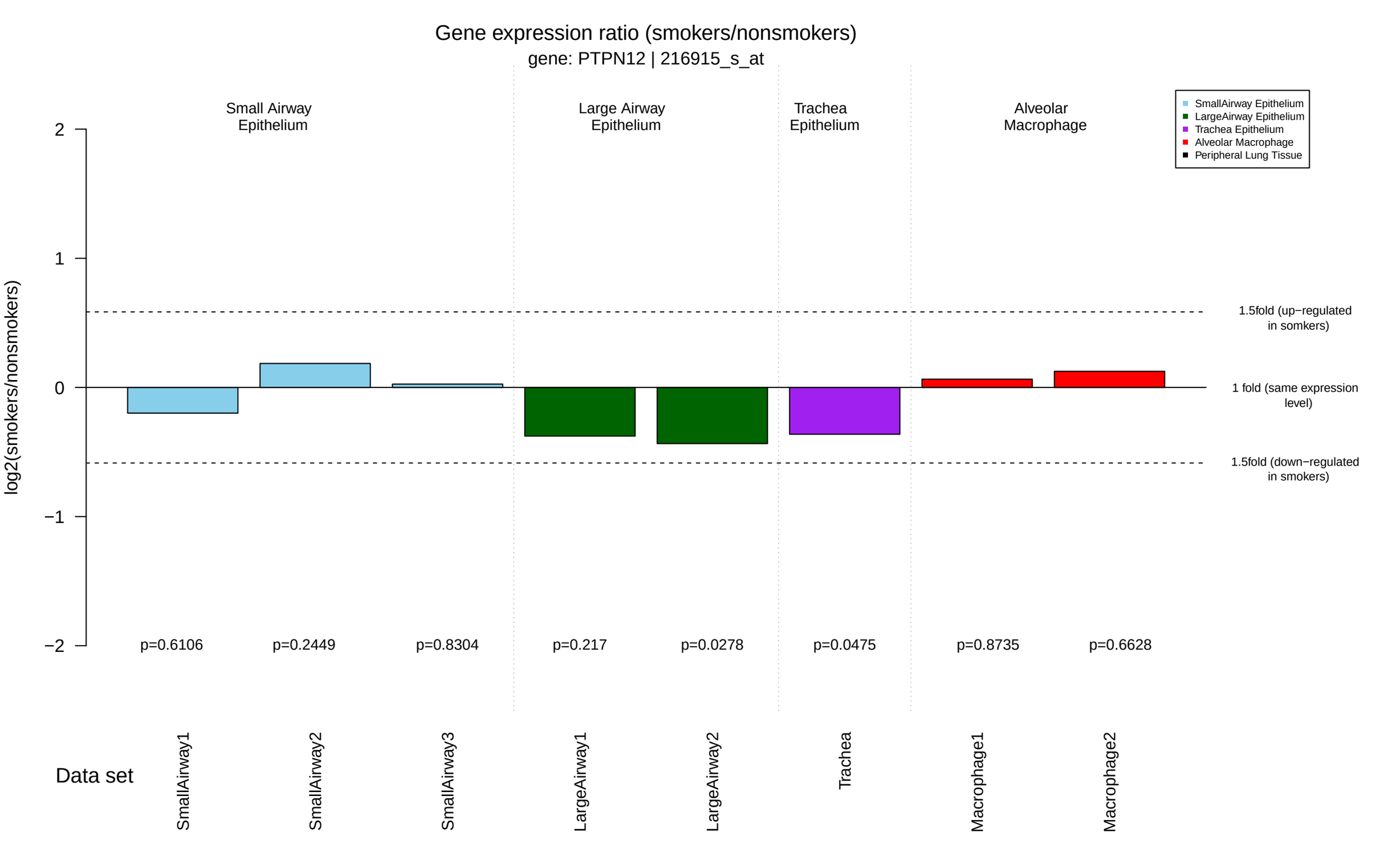
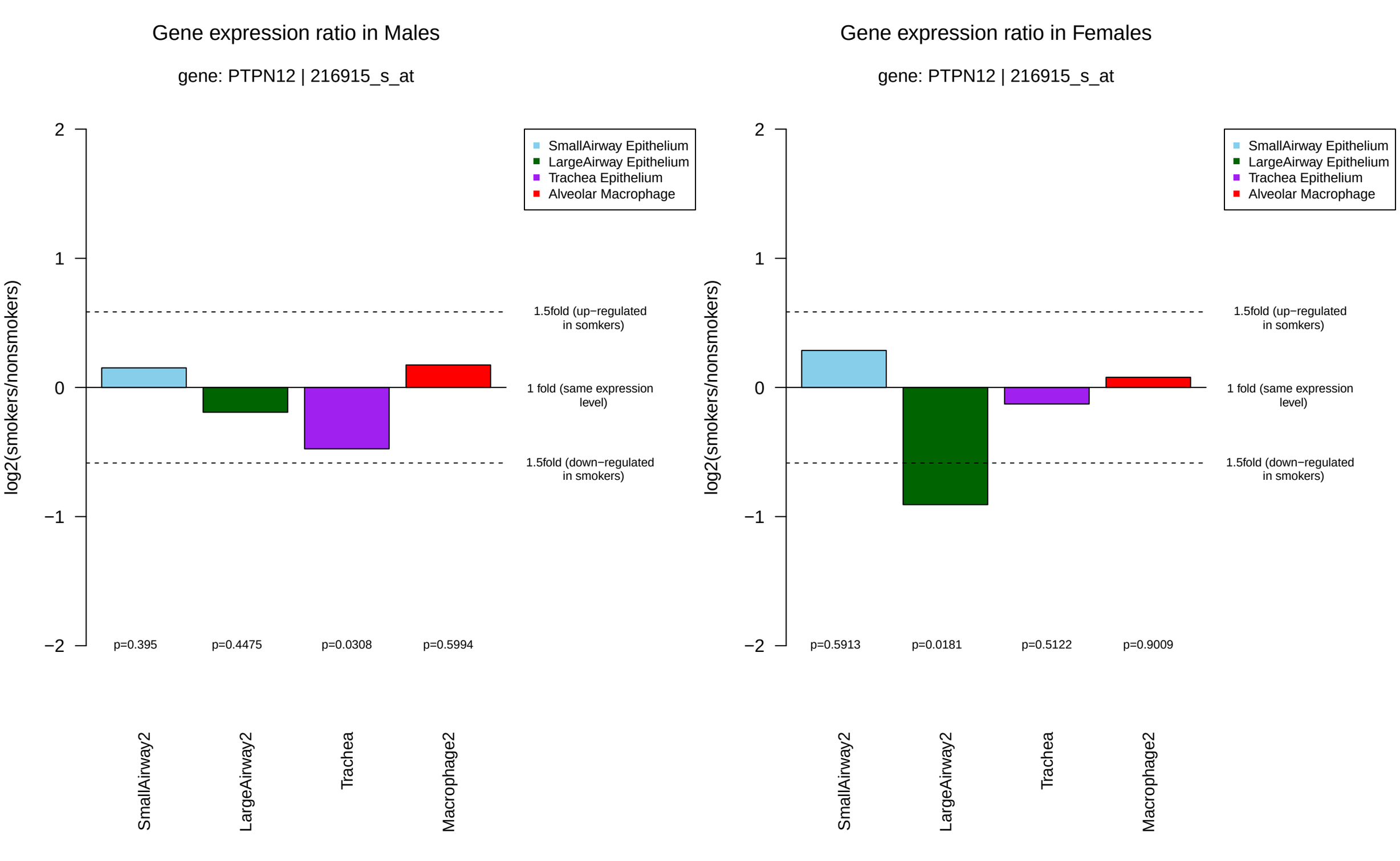
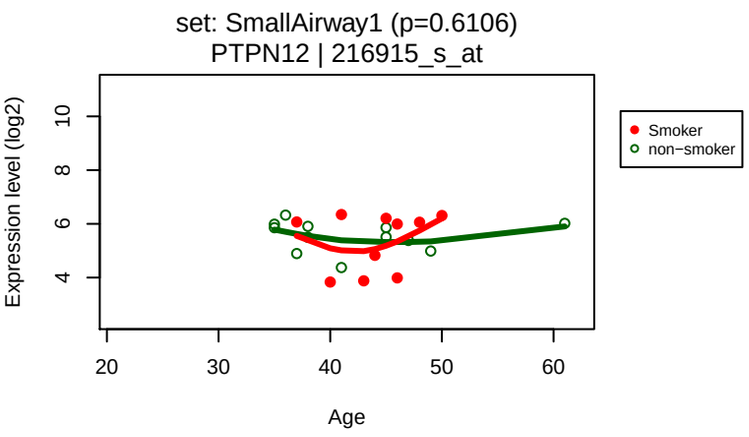
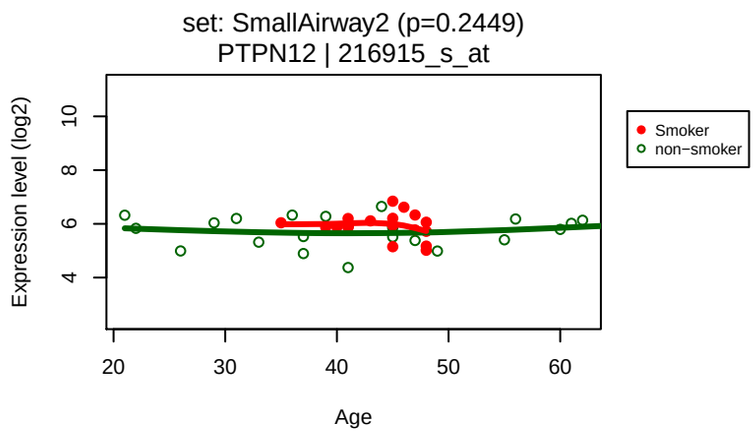
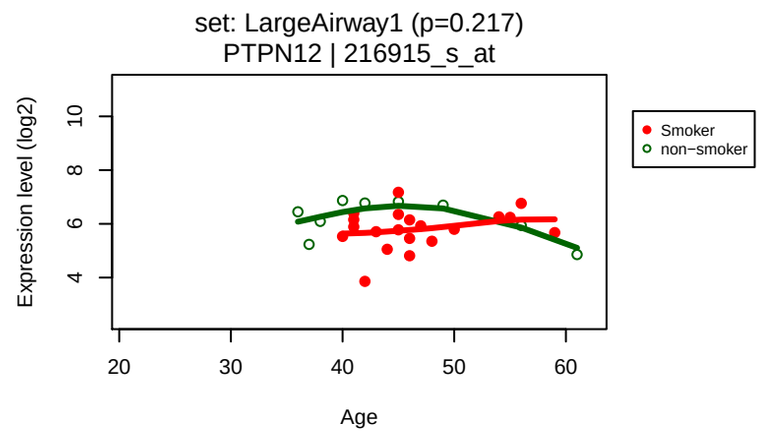

Gene expression of smokers and nonsmokers
gene: PTPN12 | 216915_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6106
p=0.2449
p=0.8304
p=0.217
p=0.0278
p=0.0475
p=0.8735
p=0.6628
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

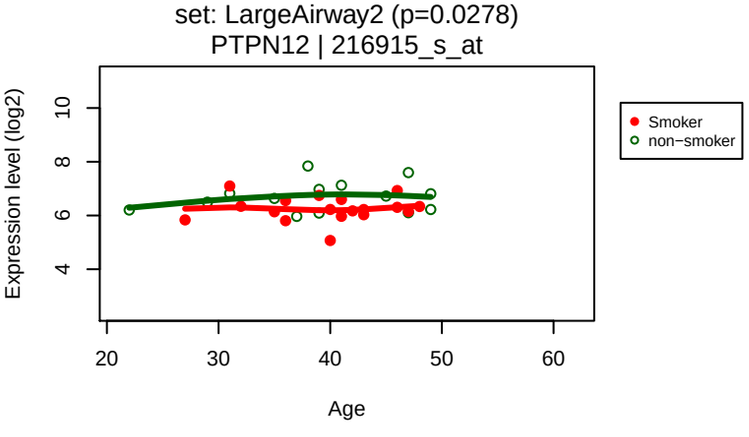

Gene expression ratio (smokers/nonsmokers)
gene: PTPN12 | 216915_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6106
p=0.2449
p=0.8304
p=0.217
p=0.0278
p=0.0475
p=0.8735
p=0.6628
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

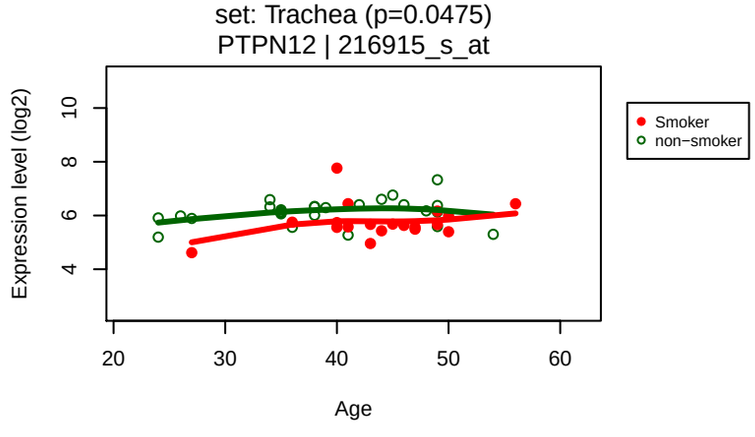

Gene expression ratio in Males
gene: PTPN12 | 216915_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.395
p=0.4475
p=0.0308
p=0.5994
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTPN12 | 216915_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5913
p=0.0181
p=0.5122
p=0.9009
SmallAirway2
LargeAirway2
Trachea
Macrophage2

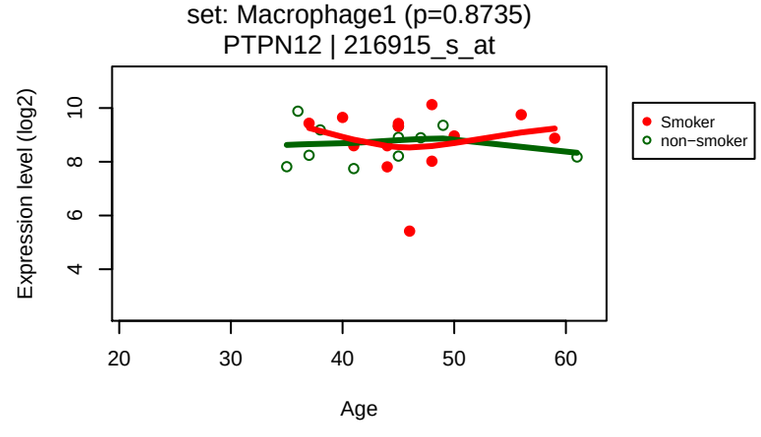

set: SmallAirway1 (p=0.6106)
PTPN12 | 216915_s_at
Expression level (log2)
Age
Smoker
non−smoker

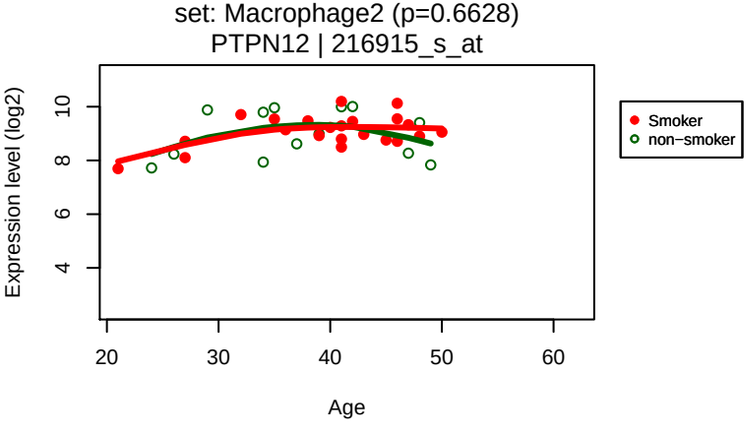

set: SmallAirway2 (p=0.2449)
PTPN12 | 216915_s_at
Expression level (log2)
Age
Smoker
non−smoker

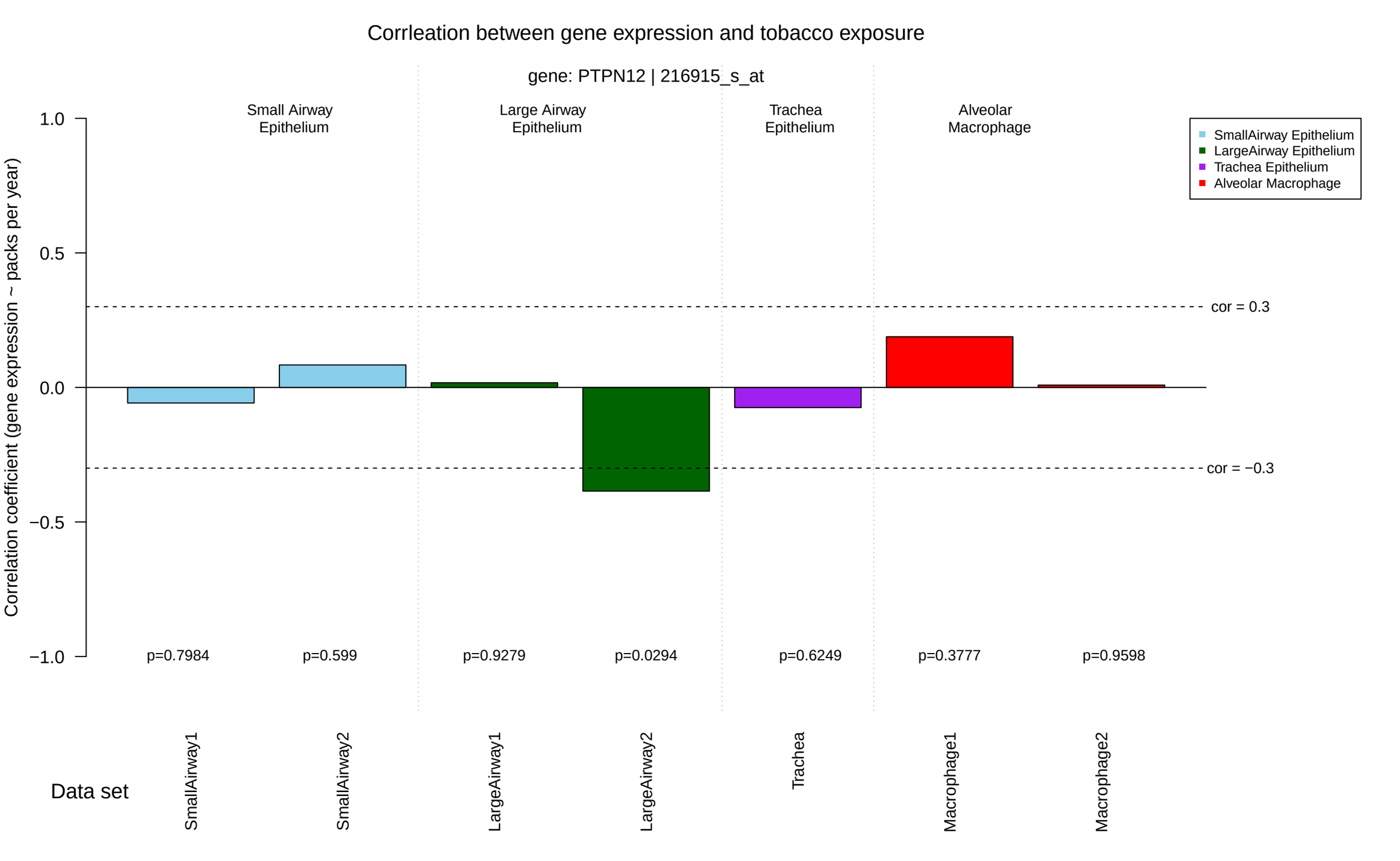

set: LargeAirway1 (p=0.217)
PTPN12 | 216915_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0278)
PTPN12 | 216915_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0475)
PTPN12 | 216915_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8735)
PTPN12 | 216915_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6628)
PTPN12 | 216915_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTPN12 | 216915_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7984
p=0.599
p=0.9279
p=0.0294
p=0.6249
p=0.3777
p=0.9598
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2